

GENERAL SPECIFICATION G5.22

Lighting System

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G5.22 LIGHTING SYSTEM

G5.22.1 General

(a) Introduction

- (i) This Section is intended to generally cover the design, supply, deliver, installation, testing and commissioning for the lighting system. The Contractor shall design the complete lighting system in compliance with SS 531 duly endorsed by Electrical PE.
- (ii) The lighting system shall comprise the luminaires, lamps, mounting structures and supports, dimming system, controls and circuit protections.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

1) International Electrotechnical Commission (IEC)

Standard	Subject
IEC 60064	Tungsten Filament Lamps for Domestic and Similar General Lighting Purposes.
IEC 60079	Electrical Apparatus for Explosive Gas Atmosphere, All Parts.
IEC 60081	Double-Capped Fluorescent Lamps – Performance Specifications.
IEC 60155	Glow-starters for fluorescent lamps.
IEC 60188	High-Pressure Mercury Vapour Lamps.
IEC 60192	Low-Pressure Sodium Vapour Lamps.
IEC 60228	Conductors of insulated cables.
IEC 60238	Edison Screw Lamp holders.
IEC 60357	Tungsten Halogen Lamps (Non-Vehicle).
IEC 60360	Standard Method of Measurement of Lamp Cap Temperature Rise.
IEC 60400	Lamp holders for Tubular Fluorescent Lamps and Starter holders.
IEC 60502	Power cables with extruded insulation and their accessories for rated voltages from 1 kV (Um = 1, 2 kV) up to 30 kV (Um = 36 kV)
IEC 60529	Degrees of Protection Provided by Enclosure (IP Code).
IEC 60598	Luminaires.

Standard	Subject
IEC 60662	High-Pressure Sodium Vapour Lamps.
IEC 60838	Miscellaneous Lamp holders.
IEC 60882	Pre-Heat Requirements for Starterless Tubular Fluorescent Lamps.
IEC 60901	Single-Capped Fluorescent Lamps.
IEC 60921	Ballasts for Tubular Fluorescent Lamps – Performance Requirements.
IEC 60923	Auxiliaries for Lamps – Ballasts for Discharge Lamps (Excluding Tubular Fluorescent Lamps) – Performance Requirements.
IEC 60924	D.C. Supplied Electronic Ballasts for Tubular Fluorescent Lamps – General and Safety Requirements.
IEC 60927	Auxiliaries for Lamps – Starting Devices (Other Than Glow Starters) – Performance Requirements.
IEC 60929	A.C. Supplied Electronic Ballasts for Tubular Fluorescent Lamps – Performance Requirements.
IEC 60968	Self-Ballasted Lamps for General Lighting Services – Safety Requirements.
IEC 60969	Self-Ballasted Lamps for General Lighting Services – Performance Requirements.
61047	D.C. or A.C. Supplied Electronic Step-Down Convertors for Filament Lamps - Performance Requirements.
IEC 61048	Auxiliaries for Lamps – Capacitors for use in Tubular Fluorescent and Other Discharge Lamp Circuits – General and Safety Requirements.
IEC 61049	Capacitors for Use in Tubular Fluorescent and Other Discharge Lamp Circuits – Performance Requirements.
IEC 61050	Transformers for Tubular Discharge Lamps having a No-Load Output Voltage Exceeding 1,000V (Generally called Neon-Transformers – General and Safety Requirements.
IEC 61167	Metal Halide Lamps.

Standard	Subject
IEC 61184	Bayonet Lamp holders.
IEC 61195	Double-Capped Fluorescent Lamps – Safety Specifications.
IEC 61199	Single-Capped Fluorescent Lamps – Safety Specifications.
IEC 61341	Method of Measurement of Centre Beam Intensity and Beam Angle(s) of Reflector Lamps.
IEC 61347	Lamp Control gear.
IEC 62560	Self-ballasted LED lamps for general lighting services with supply voltages > 50 V-Safety Specifications
IEC 62612	Self-ballasted LED lamps for general lighting services with supply voltages > 50 V-Performance Requirements
IEC 62717	LED modules for general lighting- Performance Requirements

(ii) SPRING Singapore

Standard	Subject
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.
SS 563	The Installation and Maintenance of Emergency Lighting and Power Supply Systems in Buildings.
SS 246	Fused and Unfused Adaptors.
SS 322	Earthing and Bonding Clamps.
SS 531	Code of Practice for lighting of work places.

(c) Submittals

(i) Shop Drawings, and Technical Data: As a minimum, submit the following items:

- 1) Design of lighting system in accordance with SS 531 and other relevant codes:
 - Lux level simulation and calculation using Contractor's proposed luminaires.
- 2) Interior Luminaires
 - Catalogue data sheets and pictures;

- Luminaire finish and metal gauge;
 - Lens material, pattern, and thickness;
 - Luminous intensity diagram or candle power distribution curves in two or more planes;
 - Luminous intensity diagram candle power chart 0 to 90 degrees;
 - Lumen output chart;
 - Coefficients of utilisation for zonal cavity calculations; and
 - Mounting or suspension details, and all other accessories.
- 3) Exterior Luminaires
- Catalogue data sheets and pictures;
 - Luminaire finish and metal gauge;
 - Lens material, pattern, and thickness;
 - IES lighting classification and isolux diagram;
 - Fastening details to wall or pole;
 - Ballast type, location, and method of fastening;
 - For light poles, submit wind loading, complete dimensions, and finish; and
 - Mounting or suspension details, and all other accessories.
- 4) Lamps
- Voltages;
 - Colours;
 - Approximate life (in hours);
 - Approximate initial lumens;
 - Lumen maintenance curve;
 - Lamp type and base; and
 - Copy of lamp order, including individual quantities, for Works.
- 5) Ballasts
- Type;
 - Wiring diagram;
 - Nominal watts and input watts;
 - Input voltage and power factor;
 - Starting current, line current, and restrike current values;
 - Sound rating;
 - Temperature rating;

- Efficiency ratings;
 - Low temperature characteristics; and
 - Emergency ballasts rating and capacity data.
- 6) Photo-Time Control
- Wiring diagram; and
 - Contact ratings.
- 7) Photocells
- Voltage, and power consumption;
 - Capacity;
 - Contacts and time delay;
 - Operating levels;
 - Enclosure type and dimensions; and
 - Temperature range.
- 8) Occupancy Sensors
- Type;
 - Switching capacity;
 - Coverage; and
 - Time delay AUTO/OFF adjustment.
- 9) Low Voltage Remote Control Wiring System
- Type;
 - Switching capacity;
 - Voltage rating; and
 - Wiring diagrams.
- 10) Samples
- Submit samples of each luminaire type to the S.O. for approval; and
 - Approved samples shall be tagged and left onsite to be used as a guide to the standard of all other luminaires of similar type delivered to the Site.
- 11) Operations and Maintenance Manuals
- The Contractor shall provide the as-built drawings of the complete lighting circuit (i.e. wiring connection, switches, dimmers) as part of O&M Manuals. This shall be applicable to both emergency lighting and non-emergency lighting.
- (d) Spares
- (i) Provide spare parts, tools and test equipment as specified in General Specification G2.28- Spare Parts, Tools and Test Equipment.

G5.22.2 Products

- (a) Luminaires
 - (i) All luminaire shall use LED type lamps.
 - (ii) All luminaires shall meet or exceed applicable requirements of IEC 60598 and SS 531.
 - (iii) Feed-through type, or separate junction box.
 - (iv) Ballasts / Electronic Ballasts: Two-lamp when possible meeting applicable requirements of IEC 60921. LED drivers / amplifiers shall be used where applicable.
 - (v) Earth leakage current to luminaire body not to exceed 0.5 mA.
 - (vi) Wire Leads: Minimum 1.5 mm².
 - (vii) Component Access: Accessible and replaceable without removing luminaire from ceiling.
- (viii) Soffit Installations
 - 1) Designed as SUITABLE FOR WET or DAMP LOCATIONS; and
 - 2) Ballast / Electronic Ballast: Removable, prewired.
- (ix) Exterior Installations
 - 1) Designed as SUITABLE FOR WET LOCATIONS;
 - 2) Ballast / Electronic Ballast: Removable, prewired; and
 - 3) Photocell: Where shown, factory-installed and wired, replaceable.
- (x) Marine Environments
 - 1) Designed as MARINE, OUTSIDE TYPE; and
 - 2) Housing: Copper-free, aluminium.
- (xi) Hazardous Classified Areas
 - 1) Luminaires installed in hazardous areas shall be manufactured in accordance with applicable requirements of IEC 60079, and IEC 61241; and
 - 2) Fixture Enclosure and Fittings: Copper-free, cast aluminium.
- (xii) Emergency Lighting
 - 1) Power Pack: Self-contained, dual voltage selectable input transformer, inverter/charger, sealed nickel cadmium battery, and indicator switch;
 - 2) Lighted, push-to-test indicator;
 - 3) Capable of providing full illumination for 2 hours in emergency mode;
 - 4) Capable of full recharge in 24 hours, automatically upon resumption of normal line voltage; and
 - 5) Capable of protecting against excess charging and discharging.

- (b) Lamps
 - (i) Fluorescent
 - 1) Conform to IEC 60901 and IEC 61199 for single-capped, or IEC 60081 and IEC 61195 for double-capped;
 - 2) Type Efficiency: Energy Efficient;
 - 3) Colour: White with 4,000-degree K colour temperature;
 - 4) Minimum colour rendition index of 85;
 - 5) Minimum efficacy of 90 lm/watt; and
 - 6) Guaranteed minimum life of 5,000 hours.
 - (ii) High Pressure Sodium Vapour
 - 1) Conform to IEC 60598 and IEC 60662;
 - 2) Type: Clear;
 - 3) Colour: Apparent colour temperature shall be 2,100 degrees Kelvin;
 - 4) Lumen Maintenance: Minimum 90 percent at 50 percent life;
 - 5) Guaranteed minimum life of 24,000 hours (16,000 hours for 35 watt); and
 - 6) Re-strike time shall not exceed 1 minute.
 - (iii) Metal Halide
 - 1) Conform to IEC 61167;
 - 2) Type: Coated;
 - 3) Colour: Minimum colour rendition index of 70;
 - 4) Lumen Maintenance: Minimum 60 percent at 40 percent life; and
 - 5) Guaranteed minimum life of 10,000 hours vertical, 6,000 hours horizontal.
 - (iv) Tungsten Halogen
 - 1) Type Efficiency: Energy Efficient; and
 - 2) Colour: Frosted.
 - (v) LED
 - 1) Safe deployment: Shock proof and safe for operation;
 - 2) High Efficiency: Even light distribution with no colour degradation;
 - 3) Eco-friendly: Does not contain mercury and is recyclable;
 - 4) Clean Power Lines: Low Total Harmonic Distortion (THD) (<10%);
 - 5) Operating Input Voltage: AC 220V~240V, 50Hz;
 - 6) Power Consumption: 24 watt;
 - 7) Power Factor: >0.95;
 - 8) Lumen: 1650 lm ~ 1800 lm;
 - 9) Colour: Daylight;

- 10) CRI: >70;
- 11) CCT: 4000K;
- 12) Driver: Switchable;
- 13) Operating Temp.: 0 ~ 40 degree C;
- 14) Storage Temp.: 0 ~ 40 degree C;
- 15) Dimension: approx. 1213(L) x $\varnothing$ 27mm;
- 16) Replacement For: T8 Fluorescent Tube 36W;
- 17) Type of LED: SSL Class Low Power 3020 LEDs;
- 18) Lifespan: 35,000 hrs; and
- 19) Certifications: CB, CE.

(vi) Street Light LED

- 1) Light source: up to 240 x LED;
- 2) Light colour: 4000K $\pm$ 300K;
- 3) CRI: >70;
- 4) System efficacy: 110l m/w $\pm$ 3%;
- 5) Optics: Dedicated APR Ledgine Lens;
- 6) Power requirement: 220-240V 50Hz;
- 7) Power consumption (Max): 225W $\pm$ 5%;
- 8) Driver: Constant current;
- 9) Installation Mounting: axial entry $\varnothing$ 42 – 60 mm side entry on standard pole;
- 10) Dimensions: 1050 x 400 x 90 mm;
- 11) Materials and finishing Housing: Die-cast aluminium;
- 12) Glass: Tempered glass;
- 13) Paint: Grey Paint RAL 7040;
- 14) Lifetime: 50,000 hours (70% lumen maintenance @ Ta=35°C); and
- 15) Classifications: IP 66 / IK08, Class I, Ta = 35°C.

(vii) LED Flood Light

- 1) Luminaire
 - Impact resistance;
 - IP 66; and
 - Silicon gasket.

2) Control Gear

- Aluminium gear tray, downward hinge able for easy access to components;
- Tool-less removable gear tray after disconnecting the plug;
- Class I & II safety switch standard; and
- Programmed electronic LED drivers:
 - Tuneable flux to match required lighting level within service life and luminaire configuration required;
 - Constant lumen output throughout the service life;
 - Configurable different dimming percentages and time settings; and
 - Replacement drivers are pre-mounted on a gear tray and correctly programmed.
- Cable Connection
 - Through cable gland of suitable size; and
 - Earthing of housing.
- Mounting
 - Suitable for mounting on wall / ceiling / pole as per application;
 - Adjustable angle to maximise the illumination of area; and
 - Canopy with LED module and gear tray hinges upwards and is secured by a stainless steel locking bar in two positions.
- Temperature Protection: Both LED and drivers shall have built-in protection which initially dims down and eventually switches off the light.
- LED Module
 - High quality LEDs with optimal thermal resistance and energy consumption characteristics, for high (hot) lumen output / Watt (> 125 lm / Wsys) and long expected life time;
 - Different colour temperatures available: Cool White, Neutral White and Warm White;
 - The PCB has always the same size independent of the quantity of LEDs and therefore carry always all lenses; PCBs with lower quantities of LEDs, in steps of 20, in different configurations optimized for thermal management;
 - The lenses have an optimal light transmission, in multiple blocks of 20 lenses, fixed to the PCB with two screws per lens block; fixation of the lens blocks secures equal pressure over the PCB to optimize heat spreading;

- The PCB and lenses are integrated in a high reflecting white frame to maximize light output ratio (up to 92% depending on lens type); and
 - In case of incidental LED / PCB failure, the PCB with reflector frame shall be easily replaceable after disconnecting the plug and removal of the lens blocks.
 - Glass
 - Flat glass to support the prevention of upward light, according glare classifications up to G4;
 - Toughened glass with very high light transmission, to optimize the light output ratio; and
 - Glass is fixed to die-cast aluminium frame with metal clips and can easily be replaced.
- (c) Ballasts
- (i) General: Meet requirements for fixture light output, reliable starting, radio interference, total harmonic distortion, electromagnetic interference, and dielectric rating.
 - (ii) Fluorescent
 - 1) Type: High power factor, energy efficient, electronic and instant-start type ballast, compatible with lamps specified;
 - 2) Ballasts shall be guaranteed for a service life of 10 years;
 - 3) Ballasts shall meet or exceed applicable requirements of IEC 60921 or IEC 60929 as applicable;
 - 4) The high frequency voltage generated shall be such that the frequency is well within the limits on electromagnetic interference (EMI);
 - 5) The ballast shall not generate excessive third harmonic current flow in the neutral conductor;
 - 6) Automatic resetting, thermo-protector to prevent case temperature from exceeding 110 degrees C in event of short circuit; and
 - 7) Dimming ballasts capable of operating two 36/40-watt rapid-start lamps. Include auxiliary components required for proper control.
 - (iii) Metal Halide: High power factor, normal ambient, 180 degrees C insulation class.
 - 1) Types
 - Autotransformer with capacitor and ignitor for lamps 150 watts and less; and
 - Constant wattage autotransformer with capacitor for lamps above 150 watts.

- (iv) High Pressure Sodium: High power factor, normal ambient, 180 degrees C insulation class, with capacitor and ignitor.
 - 1) Type
 - Autotransformer for 50-watt lamps; and
 - Constant wattage autotransformer for lamps 70 watts and above.
- (d) Lighting Control
 - (i) Time Switch: Provide digital electronic time switches with required number of channels. Each channel shall be independently programmable and shall have a Form C dry contact output rated for 10 amperes at 230 V AC. Time switches shall have, as a minimum, the following features:
 - 1) Selectable AM/PM or 24-hour format;
 - 2) 1-minute time resolution;
 - 3) Programmable up to 48 events per channel;
 - 4) Weekly or 365-day capability;
 - 5) Holiday capability;
 - 6) User-programmable daylight savings time adjustment option;
 - 7) Automatic leap year compensation;
 - 8) Battery backup with rechargeable batteries and 72-hour capacity; and
 - 9) Individual manual ON/OFF override control for each channel.
 - (ii) Photocell
 - 1) Automatic ON/OFF switching photo control;
 - 2) Housing: Self-contained, die-cast aluminium, unaffected by moisture, vibration, or temperature changes;
 - 3) Setting: ON at dusk and OFF at dawn;
 - 4) Time delay feature to prevent false switching; and
 - 5) Field adjustable to control operating levels.
 - (iii) Dimming System
 - 1) Fluorescent and LED dimmer capable of dimming from one to ten 36/40-watt rapid-start lamps;
 - 2) Street Light / Flood Light LED dimmer capable to dimming from one to ten 250 / 400 watt LED lamps;
 - 3) Size: Fit in a single gang wall box;
 - 4) Positive OFF switching and low intensity trim adjustment without removing dimmer from box;
 - 5) All dimming controls shall be housed in IP 55 weather proof type of enclosure for both indoor and outdoor application; and
 - 6) Zone control shall be possible from the dimming switches.

- (iv) Extra Low Voltage Remote Control Wiring System
- 1) Provide a complete extra low voltage, remote control wiring system for control of lighting fixtures. System shall be complete with all transformers, rectifiers, relays, switches, master switches, electronic controls, enclosures, wall plates, and wiring. System and all components shall be of same manufacture.
 - 2) All remote control wiring shall be in accordance with applicable requirements of CP 5.
 - 3) Provide for direct-wired connection of:
 - Standard of pilot light switches for individual control of relays; and
 - Two independent master override inputs which allow ON/OFF control of all relays while still supporting individual control of each relay.
 - 4) Relay panels shall be configured to allow future addition of up to two master controls of programmable control of all relays.
- (v) Occupancy Sensors
- 1) Passive Infrared
 - Wall switch sensors shall be capable of detection of motion at desk top level up to 27 m² and gross motion up to 92 m²;
 - Wall switch sensors shall accommodate loads from 0 to 6 amperes and shall have 180-degree coverage capability;
 - Bi-level wall switch sensors shall accommodate up to two loads from 0 to 6 amperes for each load;
 - Passive infrared sensors shall have a multiple segmented lens, in a multiple-tier configuration, with grooves-in to eliminate dust and residue build-up; and
 - Wall switches shall be compatible with electronic ballasts.
 - 2) Dual Technology Units
 - Unit to be ceiling mounted for 360-degree coverage;
 - Unit shall utilise both passive infrared and ultrasonic technologies and be easily programmed to accommodate different environmental and architectural conditions;
 - Unit must detect up to 180 m² with no blind spots; and
 - No audio dual technology units will be accepted.
 - 3) Circuit Control Hardware - CU Power Packs
 - Control Units: Able to mount through a 12 mm knock-out in a standard electrical enclosure and be an integrated, self-contained unit consisting internally of an isolated load switching control relay and a transformer to provide low voltage power. Transformer shall provide power to a minimum of two sensors; and

- Relay contacts shall have ratings of:
 - 13A, 230 V AC tungsten; and
 - 20A, 230 V AC ballast.
- 4) Wiring: Control wiring between sensors and control units shall be stranded, PVC insulated or Teflon jacketed cable approved for use in plenums, where applicable.
- 5) General
 - All sensors shall be capable of operating normally with any electronic ballast and compact fluorescent lamp systems.
 - Coverage of sensors shall remain constant after sensitivity control has been set. No automatic reduction shall occur in coverage due to cycling of air conditioner or heating fans.
 - All sensors shall have readily accessible, user adjustable controls for time delay and sensitivity.
 - In event of failure, bypass manual OVERRIDE ON key shall be provided on each sensor. When bypass is utilised, lighting shall remain on constantly or control shall divert to a wall switch until sensor is replaced. This control shall be recessed to prevent tampering.
 - Units shall have an extra Form C (1-NO-1-NC) contact for interface with building system. All units will be designed to be mountable in a standard electrical box.
 - All units shall have capability of being ordered with an integral power pack.
- (e) Poles
 - (i) Rating (With Luminaire): 160 km/hr steady winds, without incurred damage.
 - (ii) Material: Extruded aluminium or galvanised steel.
- (f) Lighting Columns
 - (i) All lighting columns shall be of mid-hinge steel pole type. The pole shall be octagonal in shape, approximately centrally hinged allowing the upper section to be lowered down through an angle of approximately 180 degrees for the lamp maintenance at ground level.
 - (ii) Lightning columns shall be manufactured from galvanised steel and designed to withstand loading comprising of wind of up to 100 km per hour, weight of the column, lantern, fitting and lamps.
 - (iii) Individual sections of the column, base plates and bracket arms shall be protected against corrosion by hot-dip galvanising internally and externally in accordance with BS EN 1461. All welding work shall be done before galvanising. The joining of two parts by welding after galvanising shall not be permitted.

- (iv) Each column shall have a base compartment adequately sized to accommodate the loop-in boxes, cut-outs and other ancillary items. The base compartment shall be of weatherproof construction fitted with a door complete with heavy duty neoprene sealing gaskets. Each door shall be provided with an 8mm Allan screw lock. The contents of the base compartment shall be mounted on a 6mm thick baseboard made of galvanised steel plate of adequate dimensions. The components shall be so arranged on the mounting plate that all components are at least 500 mm above finished ground or floor level. Rain shield shall be provided on top of the base compartment.
- (v) The base compartment of the columns shall be suitable to house a weatherproof cable loop-in box for 5 core XLPE / SWA / PVC cable complete with all accessories, terminals, glands and other accessories.
- (vi) A fully shrouded single pole and neutral single entry cut-out with a suitably rated MCB complete with all necessary accessories shall be provided for the out-going cables to the lanterns.
- (vii) All lighting columns shall be securely earthed to the cables by means of bolted earth terminals. A corrosion resistant electrical earthing terminal in the form of a bolt of not less than 6 mm in diameter and 20 mm long having substantial contact surfaces for the attachment of an earthing lead shall be provided close to the door opening of each column. The earthing shall be extended to the control gear base plate and the lanterns.
- (viii) The welded base plate of the lighting column shall be firmly bolted onto concrete block of grade 30 with minimum dimension of 500 mm (L) x 500 mm (W) x 1000 mm (H). The concrete shall be buried with 50 mm above the ground level. The four sets of bolts and nuts holding down the column shall be galvanised steel.
- (ix) Hot Dip Galvanisation for Steel Pole
 - 1) The protective coating of the steel poles shall comply with BS 5493, 'Code of Practice for Protective Coating of Iron and Steel Structures against Corrosion'. Welding and drilling shall be completed before galvanising. The entire mast shall be grit blasted before hot dip galvanised internally and externally to a minimum thickness of 85 microns (610 g/m sq). The surface shall be pickled to the requirement of BS 2451 before galvanising. The hot dip galvanizing shall comply with BS EN 1461.
 - 2) All instruments and facilities shall be provided for checking of coating thickness. All necessary precautions shall be taken to protect the galvanised surface during transportation. All defects to the coating shall be remedied at all stages of transportation, fabrication and erection. The galvaniser shall submit test reports and certificates to certify the galvanising, procedure, surface preparation, coating thickness, etc. A schedule including galvanising procedure, checking procedure, protection during transportation, handling, storage and erection shall be submitted for S.O.'s approval. A maintenance schedule shall be submitted for the galvanised coating.

- (x) Lanterns and Fittings
 - 1) All lighting lanterns and light fittings for external lighting shall be weatherproof, vandal-proof, ultra-violet (UV) stabilised, aesthetically acceptable and shall comply with IP656 construction unless otherwise approved by the S.O..
 - 2) One set of control gear comprising ballast with multiple voltage tapplings (200 - 250 volt) and power factor correction capacitor shall be supplied for each lantern/fitting.
- (xi) Control Gear Box
 - 1) A water-tight, weather proof cast aluminium box shall be provided at the base of the pole diameter, to house fuse cut-out, cable looping connects and MCB. All cable entry to this box shall be suitably sealed for water tightness.
- (xii) Cover Plate
 - 1) A one piece aluminium cover plate shall be secured to the pole to completely hide all welds, control gear box, anchor bolts, etc.
- (xiii) Labelling
 - 1) Labelling of lamp poles at a height of 1.5m above ground level with the letterings of dimension shall be provided. The Contractor shall propose the numbering, letter size and label size to the S.O. for their approval before any painting
- (g) Emergency Ballast
 - (i) In accordance with UL 924;
 - (ii) Nickel cadmium battery, charger, and electronic circuitry in metal case plus AC ballast;
 - (iii) Solid state charging indicator monitoring light and double-pole test switch;
 - (iv) Capable of operating one fluorescent lamp for a period of 120 minutes with output of 1,100 to 1,200 lumens; and
 - (v) Manufacturers: As listed in latest edition of the SPRING Singapore "Directory of SPRING Singapore Certified Products."
- (h) In-Line Fuse Holder and Fuse
 - (i) Fuse Holder
 - 1) General: Waterproof, of corrosion-resistant material; and
 - 2) Rating: 600 volts.
 - (ii) Fuse
 - 1) General: Midget, dual element; and
 - 2) Rating: 5 amperes, voltage as required by application.

G5.22.3 Execution

(a) Luminaires

(i) General

- 1) Install in accordance with manufacturer's recommendations;
- 2) Provide proper hangers, pendants, and canopies as necessary for complete installation;
- 3) Provide additional ceiling bracing, hanger supports, and other structural reinforcements to building and to concrete pole bases required to safely mount;
- 4) Install plumb and level;
- 5) Mounting heights shown for pendant mounted luminaires are measured from bottom of luminaire to finished floor or finished grade, whichever is applicable;
- 6) Mounting heights shown for wall-mounted luminaires are measured from centre of mounting plate to finished floor or finished grade, whichever is applicable; and
- 7) Install each luminaire outlet box with galvanised stud.

(ii) Pendant Mounted

- 1) Provide swivel type hangers and canopies to match luminaires, unless otherwise specified;
- 2) Space single-stem hangers on continuous-row fluorescent luminaires nominally 1,200 mm apart; and
- 3) Provide twin-stem hangers on single luminaires.

(iii) Pole Mounted

- 1) Provide cast-in-place concrete base; and
- 2) Provide branch circuit in-line fuses in pole base hand hole.

(iv) Swinging Type: Provide, at each support, safety cable capable of supporting four times vertical load from structure to luminaire.

(v) Finished Areas

- 1) Install symmetrically with tile pattern;
- 2) Locate with centrelines either on centreline of tile or on joint between adjacent tile runs;
- 3) Install recessed luminaires tight to finished surface such that no spill light will show between ceilings and sealing rings;
- 4) Install recessed luminaires tight to finished surface such that no spill light will show between ceilings and sealing rings;
- 5) Combustible Low Density Cellulose Fibreboard: Provide spacers and mount luminaires 40 mm from ceiling surface, or use fixtures suitable for mounting on low density ceilings;

- 6) Junction Boxes
 - Flush and Recessed Luminaires: Locate minimum 400 mm from luminaire; and
 - In concealed locations, install junction boxes to be accessible by removing luminaire.
- 7) Wiring and Conduit
 - Provide wiring of temperature rating required by luminaire; and
 - Provide flexible steel conduit.
- 8) Provide plaster frames when required by ceiling construction; and
- 9) Independent Supports
 - Provide each recessed fluorescent luminaire with two safety chains or two 4 mm² soft-annealed galvanised steel wires of length needed to secure luminaire to building structure independent of ceiling structure;
 - Tensile strength of chain or wire, and method of fastening to structure shall be adequate to support weight of luminaire; and
 - Fasten chain or wire to each end of luminaire.
- (vi) Unfinished Areas: Locate luminaires to avoid conflict with other building systems or blockage of luminaire light output.
 - 1) Fixture Suspension: Provide 8 mm threaded steel hanger rods. Scissor type hangers not permitted; and
 - 2) Attachment to Steel Beams: Provide flanged beam clips and straight or angled hangers.
- (vii) Building Exterior: Flush-mounted back box and concealed conduit, unless otherwise specified.
- (b) Lamps
 - (i) Provide in each fixture, number and type for which fixture is designed, unless otherwise specified.
- (c) Ballasts
 - (i) Install in accordance with manufacturer's recommendations;
 - (ii) Utilise all ballast mounting holes to fasten securely within luminaire; and
 - (iii) Replace noisy or defective ballasts.
- (d) Lighting Control
 - (i) Outdoor Luminaires: Photocells switch time clock ON at dusk with time clock switching lights OFF at pre-set time.
 - (ii) Dimming Systems
 - 1) Install in accordance with manufacturer's recommendations;

- 2) Do not connect ballasts or equipment to dimming system unless acceptable to dimming system manufacturer; and
 - 3) Provide all necessary hardware and components required for fully functional dimming system.
- (iii) Occupancy Sensors: Locate and aim sensors in correct location required for complete and proper volumetric coverage within range of coverage(s) of controlled areas per manufacturer's recommendations. Rooms shall have 90 to 100 percent coverage to completely cover controlled area to accommodate all occupancy habits of single or multiple occupants at any location within room(s). Contractor shall provide additional sensors if required to properly and completely cover respective room.
- (e) Emergency Ballast
 - (i) Install battery, charger, and electronic circuitry metal case inside fluorescent fixture housing adjacent to AC ballast;
 - (ii) Install monitoring light and double-pole switch adjacent to light fixture;
 - (iii) Connect to final circuit feeding normal lighting ahead of all local switches; and
 - (iv) Wire in accordance with manufacturer's wiring diagrams.
- (f) Emergency Lighting Unit
 - (i) Install in accordance with manufacturer's recommendations;
 - (ii) Provide permanent circuit connections with conduit and wire;
 - (iii) Connect to final circuit feeding normal lighting in area ahead of all local switches; and
 - (iv) Provide separate circuit wiring to luminaire.
- (g) Manufacturer's Services
 - (i) Occupancy Sensors
 - 1) Provide manufacturer's representative at jobsite in accordance with General Specification G2.25 – Manufacturers' Services and General Specification G2.26 – Plant Testing and Commissioning, to inspect installation, test unit, and put into service; and
 - 2) Contractor shall also provide, at Plant facility, training necessary to familiarise operating personnel with operation, use, adjustment, and problem solving diagnosis of occupancy sensing devices and systems.
- (h) Cleaning
 - (i) Remove labels and markings, except testing label;
 - (ii) Wipe luminaires inside and out to remove construction dust;
 - (iii) Clean luminaire plastic lenses with antistatic cleaners only;
 - (iv) Touch up all painted surfaces of luminaires and poles with matching paint ordered from manufacturer; and
 - (v) Replace all defective lamps at time of Substantial Completion.

G5.22.4 Appendix

- (a) APPENDIX G5.22A - Types of Luminaires.

END OF SECTION

APPENDIX G5.22A – TYPES OF LUMINAIRES

TYPE	DESCRIPTION	VOLTAGE	LAMP	MOUNTING
F4	LED FITTING, VANDAL AND CORROSION RESISTANT, IP 65, ONE-PIECE POLYCARBONATE DIFFUSER, SEAMLESS POLYURETHANE GASKET, SST TOFFLES, LED DRIVERS, NOMINAL DIMENSIONS 1266 MM(L) X 176 MM (W) X 108 MM (H)	230 V, 50 HZ	2 X 24 W LED	SURFACE / PENDANT
F5	SIMILAR TO F4 EXCEPT: C/W EMERGENCY SEALED MAINTENANCE FREE HI TEMPERATURE NI-CAD BATTERY WITH TEST SWITCH CAPABLE OF OPERATING CONTINUOUSLY FOR 2 HOURS	230 V, 50 HZ	2 X 24 W LED	SURFACE / PENDANT
F16	ZONE - 0, 1 OR 2 AS SPECIFIED, EXPLOSION PROOF LED LUMINAIRES MADE OF STAINLESS STEEL AISI-304, 0.8 MM THICKNESS. BODY MADE BY MOULDING FOR DEEP PRESSED, DONE IN ONE UNIT WITHOUT WELDING, POLYCARBONATE SURFACE OVERTUBE. NOMINAL DIMENSIONS 1347 MM(L) X 231 MM (W) X 123 MM (H)	230 V, 50 HZ	2 X 36 W LED, T8, RAPID START	SURFACE
F17	SIMILAR TO TYPE F16 EXCEPT: C/W EMERGENCY SEALED MAINTENANCE FREE HI TEMPERATURE NI-CAD BATTERY WITH TEST SWITCH CAPABLE OF OPERATING CONTINUOUSLY FOR 3 HOURS	230 V, 50 HZ	2 X 36 W LED, T8, RAPID START	SURFACE
F20	EEX D ZONE 0, 1 OR 2 AS SPECIFIED, HIGH FREQUENCY LED LUMINAIRE, STEEL FINISH WHITE STAVE ENAMELLED HOUSING WITH IMPACT RESISTANT GLASS COVER. ENTIRE ASSEMBLY TO BE RATED IP 66 MINIMUM WITH INTEGRAL HIGH FREQUENCY ELECTRONIC BALLAST. NOMINAL DIMENSIONS 682 MM (L) X 323 MM (W) X 105 MM (D)	230 V, 50 HZ	2 X 18 W LED, T8, RAPID START	SURFACE / RECESS

TYPE	DESCRIPTION	VOLTAGE	LAMP	MOUNTING
H1	HIGH BAY FITTING WITH 1.5 TO 1.7 SPACING TO MOUNTING HEIGHT RATIO. CU NOT LESS THAN 0.7, FOR ROOM INDEX OF 2 AND REFLECTANCES OF 50(C), 50(W) AND 20 (F), DIE CAST ALUMINIUM HOUSING WITH POWDER COAT FINISH, C/W DROPPED PRISMATIC BOROSILICATE GLASS REFRACTOR, SAFETY CHAIN AND BUILT-IN FUSE ASSEMBLY. NOMINAL DIMENSIONS 514 MM DIAMETER X 600 MM OVERALL HEIGHT	230 V, 50 HZ	400 W, LED	SURFACE / PENDANT
H2	SIMILAR TO H1 EXCEPT 250 WATT RATING	230 V, 50 HZ	250 W, CLEAR LED	SURFACE / PENDANT
H3	EEX D IEC ZONE 0, 1 OR 2 HAZARDOUS LOCATION AS SPECIFIED, HIGH BAY, DIE CAST ALUMINIUM HOUSING WITH POLYESTER POWDER COAT GRAY FINISH, C/W A TYPE BALLAST, TEMPERED GLASS LENS, HINGED DOOR WITH SILICONE RUBBER GASKET, ADJUSTABLE YOKE MOUNTING BRACKET, SAFETY WIRE GUARD, STAINLESS STEEL HARDWARE, SPOT DISTRIBUTION. ENTIRE ASSEMBLY TO BE RATED IP 66 MINIMUM.	230 V, 50 HZ	400 W, CLEAR LED	SURFACE / PENDANT
H4	SIMILAR TO H8 EXCEPT 400 WATT RATING			
H5	LED BASED FLOOD LIGHTING LIGHTING LUMINAIRES DIE CAST ALUMINIUM HOUSING WITH POLYESTER POWDER COAT GRAY FINISH, C/W A TYPE BALLAST, TEMPERED GLASS LENS, HINGED DOOR WITH SILICONE RUBBER GASKET, ADJUSTABLE YOKE MOUNTING BRACKET, SAFETY WIRE GUARD, STAINLESS STEEL HARDWARE, SPOT DISTRIBUTION, DIMMABLE, AND CONTROL GEAR AS PER SPECIFICATIONS. ENTIRE ASSEMBLY TO BE RATED IP 66 MINIMUM,	230 V, 50 HZ	250 W TO 400 W, LED ARRAYS TO MEET LUX LEVELS (SEE LED FLOOD LIGHT SPECIFICATIONS)	FLANGED / WALL / POLE MOUNTED

TYPE	DESCRIPTION	VOLTAGE	LAMP	MOUNTING
H6	SIMILAR TO H5 LIGHTING EXCEPT EEX D IEC ZONE 0, 1 OR 2 HAZARDOUS LOCATION AS SPECIFIED	230 V, 50 Hz	250 W TO 400 W, LED ARRAYS TO MEET LUX LEVELS (SEE LED FLOOD LIGHT SPECIFICATIONS)	FLANGED / WALL / POLE MOUNOTED
SL	ROAD LIGHTING LUMINAIRES COMPLETE WITH CLEAR ACRYLIC BOWL AND INTEGRAL CONTROL GEAR WITH DIMMING FACILITY AS A UNIT C/W 10 M GALVANIZED POLE (FLANGE MOUNTED) WITH FOUNDATION.	230 V, 50 HZ	225 W, LED (SEE STREET LIGHT LED SPECIFICATIONS)	FLANGED / POLE MOUNTED
AWL	AVIATION / OBSTRUCTION WARNING LIGHT WITH LONG LIFE LAMP, WEATHER PROOF IP 66 C/W PHOTO CELL ACCESSORIES, SOLID STATE HIGH FIX LAMP WITH MINIMUM LIFE SPAN OF 50,000 HOURS. THE HOUSING SHALL BE CAST ALUMINIUM, PAINTED. THE LENS SHALL BE OF HIGH QUALITY, UV STABILIZED ACRYLIC. THE LUMINAIRES SHALL BE OMNI-DIRECTIONAL LOW INTENSITY AVIATION / OBSTRUCTION WARNING LIGHT WHICH CONFIRMS TO ICAO / FAA / CAAS REQUIREMENTS AND RADIO INTERFERENCE – BS EN 55014 POWER CONSUMPTION TO BE LESS THAN 10 WATTS.	230 V, 50 HZ	LEDs IN 4 SEPARATE GROUPS, 6 W, TWO STAGES OVER VOLTAGE PROTECTION	ROOF TOP
EXIT	EXIT LIGHT WITH INTEGRAL HIGH TEMPERATURE LONG LIFE NICKEL CADMIUM EMERGENCY BATTERY. INJECTION MOULDED FIRE RETARDANT, CORROSION AND VANDAL RESISTANT POLYCARBONATE ENCLOSURE. ISO GREEN LEGEND. BATTERY CHARGER TO BE DUAL RATE WITH RED LED CHARGE MONITOR AND LOW BATTERY VOLTAGE CUT-OFF C/W TEST SWITCH AND ARROWS AS REQUIRED.	230 V, 50 HZ	2 X 10 W, LED	UNIVERSAL / WALL / CEILING

TYPE	DESCRIPTION	VOLTAGE	LAMP	MOUNTING
EXIT-H	IEC ZONE 0, 1, OR 2 IN ACCORDANCE WITH AREA CLASSIFICATION SPECIFIED. SEVERE ENVIRONMENT ENCLOSED EXIT SIGN WITH SEALED LEAD ACID BATTERY, CORROSION RESISTANT FIBRE GLASS HOUSING, UV STABILIZED CLEAR POLYCARBONATE COVER LED AD INDICATOR LIGHT, C/W LOW-VOLTAGE DISCONNECT AND TEST SWITCH	230 V, 50 HZ	2 X 10 W, LED	WALL / CEILING